Quantification: 872_wt_hp_z04
==============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	21
interior lacuna count	18
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 18)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1887.94	1578	1120.7	px^2
major_axis_length_px	lacuna shape	93.3239	76.683	59.7942	px
minor_axis_length_px	lacuna shape	30.4602	29.0258	10.3081	px
aspect_ratio	lacuna shape	3.3986	2.5391	2.7345	unitless
eccentricity	lacuna shape	0.8898	0.9192	0.1041	unitless
solidity	lacuna shape	0.7326	0.7575	0.0897	unitless
roots_count	headline	11.9444	11.5	6.3196	count
ring_length_r30_px	headline	497.667	474	264.072	px
ring_length_r60_px	reported	1000.28	1082.5	637.37	px
owned_length_px	ownership-dependent	1841.84	1849.98	1676.05	px
edge_count	ownership-dependent	121.667	118.5	112.207	count
mean_edge_length_px	ownership-dependent	14.6305	15.1613	4.1663	px
perimeter_px	normalised	283.011	249.797	150.102	px
ring_area_r30_px2	normalised	8663	8857	3446.5	px^2
ring_area_r60_px2	normalised	18309.7	18341.5	6863.71	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.996	1	0.0172	unitless
ring_density_r30	normalised	0.0572616	0.0552114	0.0245128	px^-1
ring_density_r60	normalised	0.0528613	0.0497017	0.0266733	px^-1
roots_per_100px_perimeter	normalised	4.4983	4.7642	2.2701	per 100 px
ring_attached_length_r30_px	network v2	467.556	453.5	261.538	px
ring_attached_length_r60_px	network v2	921.333	975.5	636.423	px
ring_length_w_r30_px	network v2	568.878	547.86	302.487	px
ring_length_w_r60_px	network v2	1140.12	1220.15	730.153	px
sholl_crossings_r10	network v2	11.2222	10.5	6.2737	count
sholl_crossings_r20	network v2	11	12.5	6.7997	count
sholl_crossings_r30	network v2	13.5	12	8.5078	count
loop_count	loops, report only	18.1111	8	23.5694	count
loop_edge_share	loops, report only	0.3567	0.3115	0.2591	share
ring_width_mean_r30_px	width	7.0933	7.1239	0.4159	px

Per field
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measure	value	units
analysed_area_px2	1.00639e+06	px^2
total_skeleton_length_px	38673	px
canalicular_length_density_per_px	0.0384274	px^-1
junction_count	1399	count
junction_density_per_px2	0.00139012	px^-2
lacuna_count	21	count

lacunae_per_px2	2.087e-05	px^-2
skeleton_component_count	144	count
mean_component_length_px	268.562	px
median_component_length_px	12	px
field_length_density_w_per_px	0.0435555	px^-1
field_density_without_flagged_per_px	0.0430498	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	80	px
unattached_component_count	6	count
canalicular_length_density_attached_per_px	0.0384274	px^-1
field_length_density_with_unattached_per_px	0.0385069	px^-1
loop_count	472	count
loop_edge_share	0.5902	share
edge_count	2745	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.6596	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.198	px
width_pixels_used	36683	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	3185	293.107	22.7028	12.9106	0.997	0.5399	-1.5651	143.896	232.375	False
2	2077	144.059	25.2241	5.7112	0.9846	0.6281	1.3769	151.947	448.976	False
3	4256	158.774	41.7686	3.8013	0.9648	0.6782	-1.4497	152.645	678.274	False
4	1429	52.067	37.8414	1.3759	0.6869	0.8279	-0.6051	154.502	593.396	False
5	1593	70.4006	31.8422	2.2109	0.8919	0.7622	1.3948	164.237	538.472	False
6	453	57.8328	13.3321	4.3379	0.9731	0.6292	1.5169	161.32	830.574	False
7	922	97.7946	15.9727	6.1226	0.9866	0.6272	1.3884	181.894	939.311	False
8	3025	100.943	44.409	2.273	0.898	0.65	-1.4641	211.119	126.158	False
9	2144	87.4647	34.0354	2.5698	0.9212	0.7528	-1.2557	226.871	777.66	False
10	2841	103.213	37.4578	2.7554	0.9318	0.8351	-1.3658	276.205	420.61	False
11	497	30.0795	23.8394	1.2618	0.6098	0.7827	-0.6524	264.441	629.968	False
12	3125	68.6713	60.8024	1.1294	0.4648	0.909	0.622	332.681	995.111	True
13	3954	114.494	48.2385	2.3735	0.9069	0.7517	-1.4049	393.248	55.5005	True
14	3573	94.8523	52.0807	1.8213	0.8358	0.8284	-1.4851	395.903	280.417	False
15	1409	76.4593	28.6406	2.6696	0.9272	0.7683	-1.5183	392.216	831.581	False
16	2226	76.9066	39.7827	1.9332	0.8558	0.7987	-1.558	506.567	185.868	False
17	1503	75.7312	26.0285	2.9095	0.9391	0.8628	-1.3051	568.045	702.943	False
18	1123	61.8357	30.8107	2.007	0.867	0.6785	-1.3397	588.985	20.1576	True
19	628	38.8866	23.9678	1.6225	0.7875	0.7235	-1.4785	619.033	332.451	False
20	1563	73.771	29.411	2.5083	0.9171	0.717	-1.5252	651.639	225.151	False
21	659	47.4878	19.9461	2.3808	0.9075	0.7744	1.5081	843.693	521.331	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	19	668	1102	1659.8	122	13.6049	6.5894
2	8	344	557	549.333	36	15.2593	6.6102
3	11	439	506	553.76	44	12.5854	7.2488
4	4	181	210	208.782	17	12.2813	7.1239
5	5	188	319	271.238	15	18.0825	7.012
6	5	207	230	92.1127	5	18.4225	7.5208
7	12	293	434	336.764	19	17.7244	6.5092
8	23	840	1399	1919.47	115	16.6911	7.2004
9	19	722	1221	1780.49	122	14.5942	7.0453
10	14	754	1743	2169.29	139	15.6064	7.206
11	9	449	1046	3143.42	192	16.372	7.8128
12	13	358	847	1304.12	86	15.1641	7.5825
13	17	607	1362	1370.58	94	14.5807	7.3359
14	17	874	2140	4594.31	305	15.0633	7.846
15	12	409	1137	2198.56	126	17.4489	6.4577
16	20	783	1870	2108.23	171	12.3288	7.3483
17	9	506	1063	2047.17	113	18.1165	6.8757
18	12	482	1240	2316.1	155	14.9426	6.8827
19	11	499	1252	6152.76	413	14.8977	6.8692
20	17	802	1761	3367.6	236	14.2695	7.3103
21	0	0	15	0	0	0	

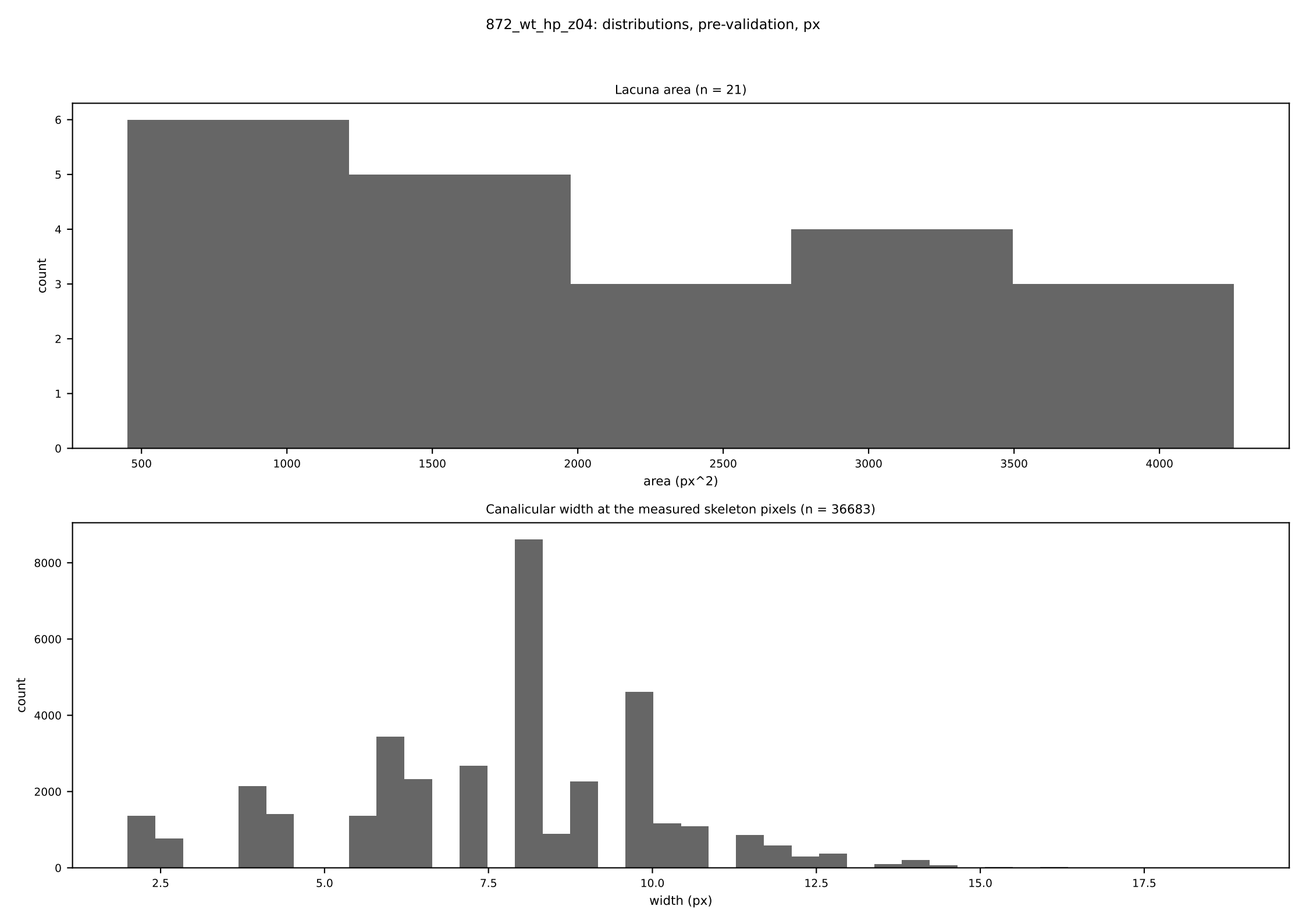
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	728.713	18655	35000	1	1	0.0358081	0.0314857	2.6073
2	364.67	8953	15872	1	1	0.0384229	0.0350932	2.1938
3	505.132	11835	18345	1	1	0.0370934	0.0275825	2.1776
4	197.238	3616	6552	1	1	0.0500553	0.0320513	2.028
5	234.723	4437	8107	1	1	0.042371	0.0393487	2.1302
6	134.361	5080	7961	1	1	0.040748	0.0288908	3.7213
7	237.723	8825	18246	1	0.9271	0.0332011	0.023786	5.0479
8	425.345	10604	20758	1	1	0.0792154	0.0673957	5.4074
9	273.321	9051	16323	1	1	0.0797702	0.0748024	6.9515
10	288.978	10581	24146	1	1	0.0712598	0.0721859	4.8447
11	117.054	5613	13620	1	1	0.0799929	0.0767988	7.6888
12	236.551	6707	15609	0.711	0.6585	0.0533771	0.0542636	5.4956
13	432.996	10444	22425	0.8508	0.7745	0.0581195	0.0607358	3.9261
14	315.191	11229	25289	1	1	0.0778342	0.0846218	5.3936
15	256.208	8359	21924	1	1	0.0489293	0.051861	4.6837
16	273.764	9314	21493	1	1	0.084067	0.0870051	7.3055
17	204.445	8382	22359	1	1	0.0603675	0.0475424	4.4022
18	216.752	5820	13951	0.7153	0.655	0.0828179	0.0888825	5.5363
19	148.61	6134	15204	1	1	0.0813499	0.0823467	7.4019
20	243.386	8889	20037	1	1	0.0902239	0.0878874	6.9848
21	145.338	6377	18338	1	1	0	0.00081797	0

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	617	1029	759.014	1243.99	20	13	11
2	259	533	388.037	639.76	6	4	10
3	430	501	509.546	588.073	11	8	9
4	181	208	206.711	238.439	3	3	4
5	162	225	216.095	361.593	5	2	5

6	188	216	237.752	262.995	6	6	3
7	261	286	328.35	486.061	7	6	7
8	809	1173	968.737	1605.09	22	15	16
9	645	1168	830.654	1403.41	16	14	21
10	732	1683	863.654	1997.43	18	19	22
11	448	1011	530.588	1196.31	9	13	11
12	308	675	399.635	964.708	7	12	12
13	564	1127	706.683	1562.71	14	14	14
14	854	2137	998.767	2464.55	19	25	31
15	342	940	458.605	1271.38	9	12	13
16	779	1853	884.441	2140.3	14	17	27
17	459	740	574.132	1183.22	11	8	15
18	475	1239	544.676	1418.12	12	12	14
19	466	1147	565.132	1416.15	10	14	20
20	784	1734	919.583	2005.28	16	19	18
21	0	0	0	18.0711	0	0	0


872_wt_hp_z04: distributions, pre-validation, px
Lacuna area (n = 21)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 36683)
count
width (px)